

Certified Contraction of Reachable Sets under Sequential Telemetry

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Abstract

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MSC 2020: [select five from the list in the source]

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1 Introduction

1.1 Static certified bounds and their blind spot

1.2 Set-membership estimation

1.3 Thesis statement

1.4 Outline of contributions

2 The Sequential Problem

2.1 Inherited formulation

2.2 Measurement epochs and the observation model

2.3 What a bounded-error return certifies

2.4 The sequential reachability problem

2.5 What contraction can and cannot mean

3 Measure Restriction and the Conditioned Problem

- 3.1 Restriction of an occupation measure
- 3.2 The conditioned Liouville problem
- 3.3 Piecewise propagation and re-initialization
- 3.4 The wrapping effect
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4 Contraction

- 4.1 Monotonicity under support restriction
- 4.2 One-shot contraction
- 4.3 Sequential contraction: the competition
- 4.4 Informativeness: when contraction is strict
- 4.5 Limits

5 Closed-Form Update of the Certificate

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5.2 The update map

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6 Verification and Results

- 6.1 Recovering the one-shot bound
- 6.2 Contraction under an informative return sequence
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7 Discussion

7.1 Limitations

7.2 Relation to observer design

7.3 Scope

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References

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D Proof of the Contraction Theorem

D.1 Monotonicity under restriction

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